Trend Following and Carry on an ETF Universe: A Pre-Registered Out-of-Sample Evaluation of the Carver Framework

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Abstract

This paper evaluates a systematic trading strategy implementing the full Carver (2015, 2019) framework — exponentially-weighted moving average crossover (EWMAC) trend rules, carry signals, volatility-scaled position sizing, a no-trade buffer, and an instrument diversification multiplier — on a universe of fourteen liquid US ETFs over the period January 2009 to October 2026. The implementation is pre-registered: the design document, signal definitions, and portfolio construction rules were all finalised before any performance calculation. The strategy achieves an annualised Sharpe ratio of 0.41 and a CAGR of 4.1%, against 0.88 for SPY, 0.99 for a daily-rebalanced 60/40, and 1.00 for a long-only book run through the same engine at a constant forecast of +10. Transaction costs of spread and stock-borrow account for approximately 0.063 Sharpe units, inside the speed limit of 0.167. Tables 1 to 4 are this book. Section 5.5 adds, without changing a rule, the T-bill on cash that the first engine omitted. With that charge the Sharpes are 0.44 and 0.87. The gap narrows from 0.59 to 0.43 and does not close. The full pipeline — 250 unit tests, expanding-window estimates, no look-ahead — is deployed in an Alpaca paper-trading loop.

1. Introduction

Systematic trend-following and carry represent two of the most extensively documented sources of risk premia in financial markets. Moskowitz, Ooi and Pedersen (2012) establish time-series momentum across asset classes; Asness, Moskowitz and Pedersen (2013) document carry; Carver (2015, 2019) provides a practitioner framework combining both in a single multi-asset system. The canonical application is futures: diverse, liquid, with clean carry signals derived from term structure curves.

A natural question is whether the same architecture can be applied to a cost-free, fractional-share ETF universe accessible to retail participants. This project addresses it in the most stringent way possible: by pre-registering the design, implementing it without look-ahead, and reporting the result as found — including the underperformance.

The contribution is not a positive result. It is a fully implemented, rigorously tested, production-quality pipeline that documents exactly *why* the strategy fails on this terrain, and deploys the identical code in a live paper-trading environment. Three results drive the analysis. First, trend-following is present but concentrated in the slowest rule (EWMAC 32/128, Sharpe 0.49 standalone) and absent in the fastest (EWMAC 8/32, Sharpe 0.05). Second, ETF carry proxies — dividend yields for equities, FRED rate differentials for bonds — are too small and too stable to function as

informative signals; standalone carry drawdowns exceed −55%. Third, removing the signals entirely, while keeping the same volatility target, allocation, and cost model, raises the Sharpe from 0.41 to 1.00. This isolates the signals as the source of underperformance and rules out explanations based on costs or execution approximations.

2. Related Literature

Trend following. Moskowitz, Ooi and Pedersen (2012) document that past twelve-month returns positively predict next-month returns across equity index, fixed income, currency and commodity futures. Baz et al. (2015) show similar results with EWMA-based filters. Hurst, Ooi and Pedersen (2017) extend the evidence to a century of data. Carver (2019) argues that multiple speeds (fast, medium, slow) each capture distinct information and that combining them via a forecast diversification multiplier (FDM) dominates any single speed.

Carry. Koijen et al. (2018) show that carry is positively related to expected returns across asset classes; the mechanism is different for equities (dividend yield minus funding cost), rates (yield spread), and commodities (futures basis). The key feature shared across asset classes is that carry is estimated from term structure information — futures curve slopes, dividend yields, rate differentials. On an ETF universe, only an approximation to the last two is available. This approximation is the source of the failure documented below.

ETF-based systematic strategies. Novy-Marx and Velikov (2016) and others note that strategy performance often degrades substantially when moved from futures to ETFs due to shorter histories, less clean carry proxies, and correlation structure dominated by equity beta.

Pre-registration in finance. Harvey, Liu and Zhu (2016) argue that the literature suffers from massive multiple testing; p-hacking inflates the reported Sharpe ratios of even honest strategies by a factor of two or more when parameters are chosen after seeing data. This project adopts pre-registration as the only structural protection against this: the design document (committed to git before any backtest run) is the single reference for all parameters.

3. Data and Universe

3.1 Price data

End-of-day prices are sourced from Tiingo. The universe consists of fourteen US ETFs representing five asset classes:

Class	Instruments	Signals
US equities	SPY, IWM	Trend + Carry
International equities	VGK, EWJ, EEM, EPP, EWC	Trend + Carry
Fixed income	IEF, TLT, TIP, LQD	Trend + Carry (TIP: trend only)
Commodities	GLD, DBC	Trend only
FX	UUP	Trend only

Selection criteria, fixed before the backtest: (i) continuous US listing since at least 2007; (ii) diversification contribution (pairwise correlation below 0.95 at selection); (iii) easy-to-borrow status

enabling short positions. SHY was excluded because its annualised volatility of approximately 1.5% produces a volatility scalar of ~138× capital, incompatible with a 2× leverage cap. TIP carry was excluded due to a structural downward bias in real yield estimates over 2011–2022.

The adjusted price series incorporates dividends multiplicatively, ensuring that return ratios and normalised signals are backward-look-through invariant. Raw (unadjusted) prices are used exclusively for position sizing and P&L attribution.

All prices are available from January 2007. The evaluation window begins 2 January 2009 (two years of warm-up for EWMA estimates, forecast scalars, and portfolio-level correlations). The final observation is 13 October 2026, yielding 4,472 evaluation days.

3.2 Rate data

Short rates (FRED DTB3, the 3-month T-bill) and bond yield proxies (FRED DGS7, DGS20, BAMLC0A0CMEY) are downloaded daily. All FRED series are lagged one calendar day before use: at date *t*, only data published before *t* is visible. This is the most consequential look-ahead control in the pipeline.

3.3 Carry proxies

For equity ETFs, carry is estimated as trailing twelve-month (252-day) dividend income per share divided by raw price, minus the T-bill rate. For IEF and TLT, carry is the respective FRED yield minus the T-bill. For LQD, carry uses BAMLC0A0CMEY, which was only added to FRED from 16 October 2023; LQD is therefore treated as trend-only for the majority of the sample, with the carry signal entering only over the final ~750 days.

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4. Methodology

The implementation follows Carver (2015, 2019) in full. The exposition here focuses on the key design choices and the look-ahead controls enforced at each step.

4.1 Volatility estimation

Daily volatility is estimated by an EWMA with span 35 (equivalent $\lambda \approx 0.943$). To avoid the well-known ghost effect of rolling windows, we use the EWMA throughout. A long-run stabiliser blends the current EWMA with its own 10-year equally-weighted mean (70/30 weight):

$$\hat{\sigma}_t = 0.70 \cdot \hat{\sigma}^{\text{EWMA}}_t + 0.30 \cdot \overline{\sigma}^{10y}_t$$

A floor equal to the 5th percentile of the blended volatility series (estimated on the past 500 days, shifted one period) prevents extreme volatility scalars during low-dispersion regimes. All estimates use only information available at the close of day *t*.

4.2 Signals

Trend (EWMAC). Three speeds are implemented: 8/32, 16/64, 32/128, denoting the span of the fast and slow exponential moving averages. The normalised signal for speed (f, s) at date t is:

$$\text{EWMAC}_{f,s,t} = \frac{\text{EMA}_{f,t} - \text{EMA}_{s,t}}{P_{t-1} \cdot \hat{\sigma}_{d,t-1}}$$

where P is the unadjusted price and σ_d the daily volatility, both shifted one period. This normalisation ensures the signal is unitless and comparable across instruments.

Carry. Two speeds are implemented: fast (EWMA span 5) and slow (EWMA span 20) of the normalised raw carry, where raw carry divided by annualised volatility is the normalised input. The EWMA filter reduces noise in the daily carry estimate.

4.3 Forecast scaling and capping

Each signal has a different natural scale. Forecast scalars (FS) bring E[|signal|] to 10 for every rule, enabling equal-weighted combination. Scalars are estimated from an expanding pooled cross-section requiring at least 100 instrument-day observations:

$$\text{FS}_t = \frac{10}{\text{pooled}\ E_t[|f|]}$$

All signal arithmetic uses the shifted (t−1) signal to produce the forecast for day t. Forecasts are capped symmetrically at ±20 after scaling. Realised forecast scalars are: EWMAC 8/32: 7.21, 16/64: 4.97, 32/128: 3.38; carry medium: ~84, carry slow: ~70. The carry scalars are large because ETF carry is small in absolute terms. After capping, E[|capped forecast|] ≈ 9.1–9.8 for all rules.

4.4 Forecast combination

Within instruments carrying both trend and carry, the combined forecast is a weighted average of the five signals. Rule weights are: 0.233 for each of the three EWMAC rules (sum 0.70) and 0.15 for each of the two carry rules (sum 0.30), following Carver's approximate 70/30 split. For instruments with no carry (GLD, DBC, UUP, TIP) weights are 1/3 each across EWMAC rules.

When a carry value is NaN on a given date (e.g., LQD before BAMLC0A0CMEY is available), the remaining rules are renormalised to sum to 1 rather than zeroing the forecast. This prevents spurious signal drop-outs.

A Forecast Diversification Multiplier (FDM) scales the combined forecast upward to restore E[|combined|] = 10, accounting for correlation between rules. The FDM is estimated on expanding pairwise correlations of the individual (scaled, capped) forecasts, with negative correlations set to zero before matrix inversion. A Herfindahl-style formula gives:

$$\text{FDM}_t = \min\!\left(\frac{1}{\sqrt{w' C_t w}}, 2.5\right)$$

where C_t is the pairwise correlation matrix. The FDM is smoothed with an EWMA of span 125. Realised values: trend+carry instruments, ~1.41 (Carver benchmark ~1.19); trend-only instruments, ~1.08 (matches benchmark). The slightly higher realised trend+carry FDM reflects the fact that ETF carry and trend have near-zero correlation, providing more diversification than in a futures universe.

4.5 Position sizing

The target position for instrument i at date t is:

$$N^*_{i,t} = \frac{\text{Capital}_t \cdot \tau \cdot F_{i,t} \cdot w_{i,t} \cdot \text{IDM}_t}{10 \cdot \sigma^{\text{annual}}_{i,t-1} \cdot P_{i,t}}$$

where τ = 13% is the portfolio volatility target (half-Kelly, consistent with an expected Sharpe of ~0.5 on a diversified book), F is the combined forecast (capped ±20), w is the instrument weight, IDM is the instrument diversification multiplier, σ_annual the annualised volatility (shifted), and P the raw close price.

Capital is the *current* portfolio value, not the initial capital. This is the key anti-ruin property: a 30% drawdown automatically reduces all positions by 30%.

Instrument weights follow a handcrafted hierarchical scheme: US equities 20% (SPY, IWM 10% each); international equities 20% (VGK, EWJ, EEM, EPP, EWC 4% each); fixed income 25% (IEF, TLT, TIP, LQD 6.25% each); commodities 25% (GLD, DBC 12.5% each); FX 10% (UUP 10%). These weights are fixed throughout.

IDM is estimated on expanding pairwise correlations of subsystem returns, where the subsystem return for instrument i is:

$$r^{\text{sub}}_{i,t} = \frac{F_{i,t-1}}{10} \cdot \frac{r_{i,t}}{\sigma_{d,i,t-1}}$$

This is the return that would be earned by instrument i if it were the only holding, scaled to unit forecast. Pairwise correlations use the same formula as the FDM. The IDM is smoothed with an EWMA of span 125, capped at 2.5, and initialised at 1. Instruments require 252 subsystem return observations before entering the portfolio; weights are renormalised during the ramp-up. Realised mean IDM over the evaluation window: 1.89.

4.6 Buffer and leverage

The gross leverage of the target portfolio frequently exceeds 2× because volatility scaling amplifies positions during low-vol regimes. Two controls are applied in sequence:

1. **Buffer.** For each instrument, define a no-trade zone: $[N - \delta|N|, N + \delta|N|]$ with $\delta = 0.10$. The held position is adjusted toward the nearest edge only when it exits the zone. The buffer absorbs small signal movements without generating trades. The buffer state is maintained in pre-leverage shares, ensuring that a leverage rescaling event does not re-trigger a trade the following day.

2. **Leverage cap.** After the buffer, gross leverage is computed. If it exceeds 2×, all positions are scaled down uniformly: scale = min(1, 2 / gross_leverage). This is a hard constraint, applied at every date.

Realised mean leverage: 1.72×. Maximum: 2.00×.

4.7 Transaction costs

Trade costs are modelled as:

$$\text{cost}_{i,t} = |\Delta N_{i,t}| \cdot P_{i,t} \cdot \text{spread} \cdot \text{slippage}$$

with spread = 2 bps (a conservative flat assumption for liquid US ETFs) and slippage multiplier = 1.5 (accounting for market impact and adverse selection). Commissions are zero (Alpaca ETF pricing). Short positions incur a daily borrow cost at instrument-specific rates (ranging from 0.3% per annum for GLD to 0.5% for IEF and TLT, following Carver's tables), charged only on negative (short) holdings.

Realised annual turnover: 24.1× capital. Sharpe cost: 0.063. Speed limit (Carver's criterion for a strategy with SR_expected = 0.5): 0.5 / 3 = 0.167. The strategy is well within the speed limit.

4.8 Benchmarks

Three benchmarks are evaluated over the identical window with the same start date:

1. **SPY buy-and-hold**: total return including dividends, zero cost.

2. **Daily 60/40**: 60% SPY + 40% IEF, rebalanced daily, zero cost.

3. **Long-only vol target**: the same engine with a constant forecast of +10 for all instruments. This benchmark shares the universe, weights, IDM, buffer, leverage cap, and transaction costs. It isolates the marginal contribution of the signals.

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5. Results

5.1 Full-sample performance

Table 1 reports annualised performance statistics over 4,472 evaluation days (2 January 2009 to 13 October 2026).

Table 1. Full-sample performance (2009–2026).

Book	Sharpe	CAGR	Volatility	Max DD	Skew	Hit rate
Strategy	0.41	4.1%	11.6%	−29.9%	−0.58	53.7%
SPY	0.88	15.2%	17.9%	−33.7%	−0.25	55.3%
60/40	0.99	10.2%	10.4%	−21.0%	−0.14	55.7%
Long-only vol target	1.00	10.1%	10.2%	−19.8%	−0.50	55.4%

The strategy achieves a lower Sharpe ratio than all three benchmarks. The relevant comparison for signal evaluation is the long-only vol target: both books have essentially the same volatility (~11–12% with leverage, ~10% after the cap) and the same costs. The Sharpe gap of 0.59 is attributable solely to the forecasts.

Leverage dynamics: mean 1.72×, maximum 2.00× (the cap binds regularly during low-volatility regimes). Turnover: 24.1×. Cost drag in Sharpe terms: 0.063 vs. the long-only vol target's 0.006. Even attributing the full incremental cost (0.057 Sharpe units) to signals, the unexplained gap is 0.53.

The $100,000 account started in January 2007, warmup included, ends near $226,000. The 4.1% CAGR belongs to the 2009–2026 window only, and compounds to about $202,000. The two figures are not the same path.

5.2 Walk-forward decomposition

To assess whether the full-sample result masks structural breaks, four non-overlapping periods are evaluated without any re-fitting of parameters. Table 2 reports results.

Table 2. Walk-forward Sharpe ratios (no parameter updates).

Period	Sharpe	CAGR	Max DD
2009–2011	0.45	5.2%	−10.1%
2012–2016	0.41	3.6%	−12.3%
2017–2021	0.77	8.2%	−11.7%
2022–2026	0.06	−0.1%	−29.9%

Three of four periods produce a positive Sharpe. The block labelled 2022–2026 contains the published maximum drawdown of −29.9%. It is one block, not a year. Section 5.5 splits it. The label "2009–2012" in the code ends on 31 December 2011. No parameter is refit between blocks, so these are subperiods.

5.3 Rule-level decomposition

Each rule is evaluated by running the full engine on that rule's capped forecast alone. The Sharpe is the Sharpe of that book. It is not the marginal effect of adding the rule to the combination.

Table 3. Standalone rule performance.

Rule	Sharpe	Turnover	Max DD	Sharpe cost
EWMAC 8/32	0.05	44.1×	−28.1%	0.106
EWMAC 16/64	0.34	22.7×	−28.7%	0.056
EWMAC 32/128	0.49	12.7×	−29.8%	0.032
Carry medium	0.18	3.7×	−56.8%	0.011
Carry slow	0.17	2.6×	−58.7%	0.008

Three observations:

1. **Trend is monotonically better at longer horizons.** The fast rule (8/32) is essentially flat (Sharpe 0.05) and its cost in Sharpe units (0.106) nearly equals its gross alpha. The slow rule (32/128, Sharpe 0.49) is the primary source of trend premium.

2. **Carry drawdowns are extreme.** Standalone carry maximum drawdowns exceed −55%, an order of magnitude larger than what the carry rules contribute positively to the combined book. ETF carry is too stable and too mean-reverting to function as a reliable signal over shorter horizons.

3. **The combined book is below the best single rule.** The combined Sharpe of 0.41 sits below the standalone EWMAC 32/128 at 0.49. The difference is not the cost of carry. Carry's own books, in this table, have Sharpes of 0.18 and 0.17 and drawdowns past −55%. The slow trend, alone, is already 0.51 below the long-only control.

5.4 Asset-class contribution

Table 4 reports the Sharpe ratio of each asset class's daily P&L contribution to the portfolio, expressed as a fraction of starting capital.

Table 4. Asset-class contribution Sharpe ratios.

Class	Sharpe
Commodities	0.45
Fixed income	0.35
FX	0.19
US equities	0.14
International equities	−0.09

Commodities and fixed income drive positive attribution; international equities are mildly negative. These are shares of the portfolio P&L, not standalone sleeves.

5.5 Cash financing, added after the tables

Tables 1 to 4 do not charge the T-bill on cash. A long position above the account's equity was borrowing for free. Short proceeds earned nothing. The rate was already in the data, lagged one day, and used by the carry rule. Putting it on the cash balance changes no forecast, no weight, and no volatility target. The sample in this section runs to 23 October 2026, ten days past Table 1. That is not what moves the Sharpes.

Cash is equity minus the marked value of the positions. A negative balance pays the lagged three-month bill. A positive balance, which is what a short leaves behind, earns it. The excess Sharpe then subtracts the same bill from the daily return, so a book that only holds cash scores zero. SPY and the 60/40 are unlevered. Their total-return Sharpe is the one in Table 1. Only the excess column charges them for the bill.

Table 5. Same books after the cash charge.

Book	Total Sharpe	Excess Sharpe	CAGR	Volatility	Max DD
Strategy	0.44	0.31	4.5%	11.5%	−25.1%
Long-only vol target	0.87	0.73	8.7%	10.2%	−20.3%
SPY	0.87	0.80	15.0%	17.8%	−33.7%
60/40	0.99	0.85	10.2%	10.4%	−21.0%

The control falls from 1.00 to 0.87. It was the book that borrowed and did not pay. The strategy moves from 0.41 to 0.44. The gap narrows from 0.59 to 0.43. In excess of the bill the gap is 0.73 − 0.31. On a financed slow-trend book the Sharpe is 0.48, and on the two financed carry books it is 0.40 and 0.41. The slow trend remains far from the control. Carry is not the gap of Table 1, and it is not the gap of Table 5.

Table 6 is a presentation cut, not a new estimation. It replaces the habit of reading the fourth block as "2022".

Table 6. Annual total return of the financed book.

Year	Strategy	Long-only	SPY
2009	2.3%	15.2%	26.4%
2010	8.7%	21.2%	15.1%
2011	4.4%	9.3%	1.9%
2012	−0.9%	12.0%	16.0%
2013	6.9%	−2.6%	32.3%
2014	10.5%	6.0%	13.5%
2015	−4.4%	−8.7%	1.3%
2016	6.4%	14.0%	12.0%
2017	6.2%	15.6%	21.7%
2018	4.8%	−7.5%	−4.6%
2019	7.5%	26.5%	31.2%
2020	15.1%	9.4%	18.4%
2021	6.5%	14.1%	28.7%
2022	7.6%	−14.0%	−18.2%
2023	−8.1%	9.4%	26.2%
2024	−5.5%	12.0%	24.9%
2025	9.6%	20.7%	17.7%
2026	3.7%	9.2%	12.3%

2022 is +7.6% for the strategy and −14.0% for the long-only book. The financed maximum drawdown is −25.1%, on 27 January 2025. The losing years are 2023 and 2024.

From 2023 onward the slow equity-carry forecast is negative on essentially every day for SPY, IWM, VGK, EWJ, EEM, EPP and EWC. Dividend yield was below the bill, so the rule was short equities. Inside the combined book, 2023 is negative in every asset class: commodities −4.5%, international equities −3.8%, US equities −2.9%, FX −1.4%, bonds −0.7%. In 2024 international equities are −4.0%, bonds −3.5% and US equities −3.0%, while commodities are +3.0% and FX +3.1%. Standalone, the slow trend makes +15.9% in 2022, −11.1% in 2023 and +6.0% in 2024. The slow carry makes −15.4%, −6.0% and −9.3%. The 2023 loss is mostly the slow trend. The 2024 loss is mostly carry.

No paired test is attached to either gap. Over about eighteen years the standard error of an annual Sharpe is about 0.24.

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6. Discussion

6.1 Why the signals underperform on ETFs

The long-only vol target in Table 1 has a Sharpe of 1.00. The strategy has 0.41. The difference is the forecasts, on a book that does not yet charge the T-bill for borrowed cash. Section 5.5 charges it. Equity carry in this implementation is the trailing dividend yield minus the T-bill, which is also, roughly, the carry of an equity-index future. Bond carry uses a daily FRED yield and has no roll-down. Commodity and FX carry are absent. A slow trend book at 0.49 does not need a weak carry proxy to sit far below the control.

The 2009–2026 equity bull market punishes short equity signals. The EWMAC framework generates short positions when prices trend down and holds them until the trend reverses. US equities in this sample experienced three brief drawdowns (2009, 2020, 2022) interspersed with strong upward trends. Short signals generated in drawdowns were frequently reversed before they could profit. International equities underperformed US equities throughout, generating additional short signals against the global equity bid.

6.2 What the framework does correctly

Despite the signal underperformance, several design elements function as intended:

- **Volatility targeting**: the strategy stays close to its 13% target except in high-leverage, low-volatility regimes where the 2× cap binds. Mean leverage of 1.72× indicates that the strategy is not systematically over-sized.
- **The buffer works**: approximately 29% of share variation is absorbed by the buffer; only 66% of instrument-days generate a trade. Turnover of 24.1× is substantially below the speed limit.
- **The long-only vol target is competitive**: Sharpe 1.00 vs. 0.99 for the 60/40, with materially lower maximum drawdown (−19.8% vs. −21.0%). The allocation and sizing engine, stripped of signals, is a credible portfolio.
- **The first three blocks are positive.** Their Sharpes are 0.45, 0.41 and 0.77. The fourth block is where the published drawdown sits. Which year inside that block produces it is measured in Section 5.5, on the financed book.

6.3 No-tuning commitment

No parameter was revised after the backtest was run. The pre-registered design document was committed to version control before any performance calculation. This constraint eliminates the most common source of illusory backtested performance: in-sample parameter optimisation. The documented Sharpe of 0.41 is a genuine out-of-sample statistic under the Carver framework, not an upper bound on what this methodology can achieve in-sample.

The 250-test suite covering every module from volatility estimation through buffer computation ensures that the implementation is correct. No test was written after a failure was discovered post-facto; tests define the contract of each function.

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7. Live Paper Trading

The same codebase is deployed in a paper-trading loop on Alpaca. The loop runs every five minutes, determining whether to:

1. *Close* (after the NYSE session ends, +20 minutes): update prices and rates, recompute the full book, write target positions and orders to SQLite — nothing is submitted.

2. *Open* (at the NYSE open): submit limit orders, passive on the bid (buys) or ask (sells).

3. *Work* (during the session): re-peg unfilled limits every five minutes; convert to market after 30 minutes.

Three live-specific risk controls are added: a circuit-breaker that suspends order submission if the day's P&L falls below −5% of equity; a preventive leverage gate at 1.5× (blocks exposure-increasing orders while allowing reductions); and an ADV filter that clips orders to 1% of average daily dollar volume.

The broker is accessed through an abstract `BrokerInterface`. The current implementation (`AlpacaBroker`) targets the paper endpoint; a future `IBKRBroker` would require no changes to the strategy layer.

The audit trail records, for every session: target positions, forecast values, weights, IDM, held positions at the broker, orders planned versus submitted versus filled, and any risk override. This constitutes the data required to attribute live deviations from backtest to execution (slippage, partial fills, data latency) versus strategy drift.

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8. Conclusion

This project implements the Carver systematic trading framework from scratch, in production-quality Python, on a fourteen-ETF universe, and evaluates it over 17 years without re-fitting any parameter after seeing performance. On this sample the signals trail a long-only book in the same engine. After charging the T-bill on cash, that gap is about 0.43 of Sharpe, and it is already present in the slow trend alone. The 2023–2024 losses, not 2022, produce the drawdown. A paired test of the gap is not reported.

The contribution is threefold: a rigorous implementation that can serve as a foundation for a futures-based V2; a methodologically clean control experiment that isolates signals from allocation; and a live paper-trading deployment that closes the loop from research to execution.

For a V2 using futures, the infrastructure is in place. The `BrokerInterface` abstraction, the carry module (which needs only a different data source), and the buffer and sizing layers are all futures-agnostic. The primary addition required is a contract roll module and an IBKR broker implementation.

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Code repository available on request. The design document, unit tests (250), and backtest engine are committed to version control with pre-registration semantics: no parameter in the codebase was modified after the first performance calculation.